Ziwei Wang

Ph.D. Candidate in Intelligent Science and Technology

vivi@hust.edu.cn | wzwvv.github.io | wzwvv | Scholar

Profile

I develop data-efficient, robust, and generalizable learning methods for neural signals under limited labels, strong heterogeneity, and distribution shifts. My work spans cross-species and cross-modality transfer, interpretable EEG decoding, multimodal learning, and physiologically grounded data generation. My long-term goal is to build trustworthy cross-species, multimodal, and multitask foundation models for brain-disorder screening, prognosis, treatment-response monitoring, and rehabilitation.

>97%	20+	2,000+
Cross-Species ESD AUC	Generated Datasets	Open-Dataset Downloads

Education

Huazhong University of Science and Technology — Wuhan, China
Ph.D. in Intelligent Science and Technology; Advisor: Prof. Dongrui Wu — 2023 – Present

Prof. Wu is an NSFC Distinguished Young Scholar, IEEE Fellow, and Editor-in-Chief of *IEEE Transactions on Fuzzy Systems*.

Huazhong University of Science and Technology — Wuhan, China
M.S. in Intelligent Science and Technology — 2021 – 2023

Central South University — Changsha, China
B.Eng. in Measurement and Control Technology and Instrumentation — 2017 – 2021

Research Vision

- **Cross-domain generalization:** transfer across subjects, datasets, devices, modalities, and species.
- **Generalizable neural decoders:** interpretable spatiotemporal, frequency-aware, contrastive, and multimodal learning.
- **Foundation-model data:** knowledge-guided augmentation, generation, denoising, benchmarking, and trustworthy evaluation.

Research Funding

- **Principal Investigator**, NSFC Young Student Basic Research Program, Grant 625B2077, "Cross-Species Multimodal EEG General Representation and Foundation Models," RMB 300,000, 2026–2027.
- **Principal Investigator**, CAST Young Elite Scientists Sponsorship Program, Doctoral Student Track, "General-Purpose EEG Foundation Models," RMB 40,000, 2026–2027.

Selected Honors

- **Ant Group InTech Scholarship**, one of 10 recipients worldwide, 2025.
- **National Scholarship**, four-time recipient: Ph.D. 2024–2026; undergraduate 2020.
- **Excellent Scientific and Technical Paper**, Hubei Association for Science and Technology, 2025; **Merit Student Pacesetter**, Highest Student Honor in HUST, 2025.
- **IEEE Computational Intelligence Society Student Grant**, one of five recipients worldwide, 2022.
- **Top 10 Advances in Brain–Computer Interfaces in China**, 2025; **Top 10 Highlights**, 2024; **National First Prize**, World Robot Contest BCI Competition, 2023.

Selected Publications

*Equal contribution #Corresponding author Complete publication list

Cross-Species and Cross-Domain Neural Decoding

1. **Canine EEG Helps Human: Cross-Species and Cross-Modality Epileptic Seizure Detection via Multi-Space Alignment.**
   Ziwei Wang, Siyang Li, and Dongrui Wu#. **National Science Review, 2025**
   The first systematic demonstration that heterogeneous canine/human and intracranial/scalp EEG can improve cross-species seizure detection; achieved over 90% AUC with very limited target labels. [Code] [Data]

2. **CKD: Contrastive Knowledge Distillation for Cross-Dataset EEG Classification.**
   Ziwei Wang, Xingyi He, Hongbin Wang, and Dongrui Wu#. **IEEE Transactions on Biomedical Engineering, 2026**
   A two-stage framework combining cross-dataset teacher pretraining with cross-subject online adaptation through logit distillation and feature-level contrastive alignment. [Paper]

3. **Unsupervised Domain Adaptation for Cross-Patient Seizure Classification.**
   Ziwei Wang, Wen Zhang, Siyang Li, Xinru Chen, and Dongrui Wu#. **Journal of Neural Engineering, 2023**
   A multi-source adaptation framework integrating source selection, global alignment, and target-agent subdomain adaptation for new patients without target labels. [Code]

Generalizable, Interpretable, and Multimodal Neural Decoders

4. **DBConformer: Dual-Branch Convolutional Transformer for EEG Decoding.**
   Ziwei Wang, Hongbin Wang, Tianwang Jia, Xingyi He, Siyang Li, and Dongrui Wu#. **IEEE Journal of Biomedical and Health Informatics, 2026**
   Parallel temporal and spatial Conformer branches model long-range dynamics and channel interactions across three EEG paradigms, with over eight-fold fewer parameters than high-capacity Conformer baselines. [Code]

5. **MVCNet: Multi-View Contrastive Network for Motor Imagery Classification.**
   Ziwei Wang, Siyang Li, Xiaoqing Chen, and Dongrui Wu#. **Knowledge-Based Systems, 2025**
   Time-, frequency-, and spatial-view augmentation is coupled with cross-view and cross-model contrastive learning, consistently outperforming nine baselines on five motor-imagery datasets. [Code]

6. **FAConformer: Frequency-Aware Convolutional Transformer for Auditory Attention Decoding.**
   Ziwei Wang*, Xingyi He*, Tianwang Jia, Hongbin Wang, and Dongrui Wu#. **Science China Information Sciences, under review, 2026**
   Band-specific encoders, frequency-aware attention, and auxiliary supervision improve auditory attention decoding across two datasets and three decision-window lengths. [Code]

7. **RAMamba-Net: A Reliability-Aware and Mamba-Based Multimodal Fusion Network for Auditory Attention Detection.**
   Xingyi He*, **Ziwei Wang***, and Dongrui Wu#. **ICLR 2027, under review**
   Reliability-aware EEG–EOG fusion combines band-aware Conformer and Mamba encoders, cross-modal attention, and sample-wise modality weighting. [Preprint]

Selected Publications

*Equal contribution #Corresponding author Complete publication list

Data Generation and Foundations for Large Neural Models

8. **Time-Frequency Transform Based EEG Data Augmentation for Brain–Computer Interfaces.**
Ziwei Wang, Siyang Li, Xiaoqing Chen, and Dongrui Wu#. **Knowledge-Based Systems, 2025**
DWTaug and HHTaug jointly address data scarcity, non-stationarity, and inter-subject variability; evaluated on 17 datasets from three BCI paradigms. [Code]

9. **Channel Reflection: Knowledge-Driven Data Augmentation for EEG-Based Brain–Computer Interfaces.**
Ziwei Wang*, Siyang Li*, Jingwei Luo, Jiajing Liu, and Dongrui Wu#. **Neural Networks, 2024**
A parameter-free augmentation method that converts paradigm-specific spatial priors into channel mappings for data-efficient and interpretable EEG decoding. [Code]

10. **Synthetic Data Generation for Brain–Computer Interfaces: Overview, Benchmarking, and Future Directions.**
Ziwei Wang, Zhentao He, Xingyi He, Hongbin Wang, Tianwang Jia, Jingwei Luo, Siyang Li, Xiaoqing Chen, and Dongrui Wu#. **Artificial Intelligence Review, minor revision, 2026**
A unified taxonomy and benchmark covering four BCI paradigms and four evaluation dimensions: signal realism, physiological plausibility, downstream utility, and privacy. [Code]

11. **Exploring Diffusion Transformers for Cross-Modal Augmentation in Multimodal Brain State Decoding.**
Ziwei Wang, Xingyi He, Hongbin Wang, Tianwang Jia, Bohan Fang, and Dongrui Wu#. **ICLR 2027, under review**
CoMA-DiT treats paired modalities as mutual generative supervision through cross-modal attention and reliability-gated latent augmentation. [Preprint]

12. **DenoFlow: Flow Matching for SSVEP Denoising under Real Physiological Artifacts.**
Zhentao He*, **Ziwei Wang***, and Dongrui Wu#. **Knowledge-Based Systems, minor revision, 2026**
A rectified-flow denoising framework trained with recorded EMG/EOG artifacts and task-aware supervision, improving both signal fidelity and downstream SSVEP decoding. [Code]

Selected Collaborative Publications

1. **Purity-Guided Active Prompting for EEG Decoding with Large Language Models.**
Jingwei Luo, **Ziwei Wang**, Dingkun Liu, and Dongrui Wu#. **IEEE Computational Intelligence Magazine, 2026**

2. **Black-Box Test-Time Ensemble.**
Siyang Li, **Ziwei Wang**, Cong Liu, and Dongrui Wu#. **IEEE Computational Intelligence Magazine, 2026**

3. **Alignment-Based Adversarial Training for Improving the Robustness and Accuracy of EEG-Based BCIs.**
Xiaoqing Chen, **Ziwei Wang**, and Dongrui Wu#. **IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2024**

4. **T-TIME: Test-Time Information Maximization Ensemble for Plug-and-Play BCIs.**
Siyang Li, **Ziwei Wang**, and Dongrui Wu#. **IEEE Transactions on Biomedical Engineering, 2023**

5. **Multi-Source Decentralized Transfer for Privacy-Preserving BCIs.**
Wen Zhang, **Ziwei Wang**, and Dongrui Wu#. **IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2022**

Industry Experience

Alibaba Cloud China
Algorithm Research Intern, Cloud-Server Failure Prediction Oct. 2022-Apr. 2023

- Proposed a SparseAttention module to improve the efficiency and predictive performance of long-sequence forecasting.
- Designed a domain-generalized mixture-of-experts architecture for robust fault prediction under temporal shifts and device heterogeneity.
- Developed five anomaly-aware augmentation operators based on N-grams and TF–IDF.

Selected Patents

1. Dongrui Wu and **Ziwei Wang**. **Cross-Subject Target-Domain Proxy Subdomain Adaptation Method, System, and Medium**. Chinese Invention Patent ZL202211350636.X, granted. *First student contributor*.
2. Dongrui Wu, **Ziwei Wang**, et al. **Supervised Contrastive Learning Method and Device Based on Spatiotemporal-Frequency Consistency**. Chinese Invention Patent ZL202410609125.8, granted. *First student contributor*.
3. Dongrui Wu and **Ziwei Wang**. **Contrastive Knowledge Distillation for Cross-Dataset Motor-Imagery EEG Decoding**. Chinese Invention Patent CN122332960A, published. *First student contributor*.
4. Dongrui Wu, **Ziwei Wang**, et al. **EEG Denoising Model Training, Denoising, and Data Augmentation Methods**. Chinese Invention Patent CN122262473A, published. *First student contributor*.

Selected Presentations

- **Tutorial**, "Transfer Learning for EEG-Based Brain–Computer Interfaces," ICONIP 2023, Changsha, China.
- **Oral**, "Cross-Species and Cross-Modality Epileptic Seizure Detection via Multi-Space Alignment," China Systems Science Congress 2025, Kunming, China.
- **AI for Science Showcase**, "EEG-Based Automatic Seizure Detection," Alibaba Cloud Yunqi Conference 2024, Hangzhou, China.
- **Posters**, China Brain–Computer Intelligence Conference 2024; SAAC 2024 and 2025.

Academic Service

Reviewer for ***Artificial Intelligence Review (IF=18.8)***, ***IEEE Reviews in Biomedical Engineering (IF=16.0)***, *IEEE Transactions on Fuzzy Systems*, *IEEE Journal of Biomedical and Health Informatics*, *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *IEEE Transactions on Biometrics, Behavior, and Identity Science*, *Knowledge-Based Systems*, *Neurocomputing*, *Journal of Neural Engineering*, *Scientific Reports*, ICONIP, and IEEE SMC. Recognized as an **IOP Trusted Reviewer** in 2024.

Open Science

Released the **CHSZ pediatric epilepsy EEG dataset**, collected with Wuhan Children's Hospital from 27 children with epilepsy. The dataset has received more than 2,000 downloads. Source code for representative projects is available at **github.com/wzwvv**.